

CS 231: Data Structures and Algorithms

Spring Semester — 4 Credits

Instructor: Dr. A. Whitfield | Office Hours: Tue/Thu 2-3pm, Room 214

Course Description

This course covers fundamental data structures and algorithms, including lists, stacks, queues, trees, hash tables, and graphs, along with techniques for analyzing time and space complexity. Students will implement each structure from scratch and apply them to solve computational problems of increasing difficulty.

Grading Breakdown

Component	Weight
Problem Sets (3)	15%
Midterm Exam	25%
Final Project	30%
Quizzes	10%
Class Participation	20%

Assignments and Exams

Problem Set 1 (5% of course grade) is due **Week 3 Friday**. It covers asymptotic analysis and Big-O notation.

Problem Set 2 (5%) covers linked lists and stacks, and is due **Week 6 Friday**.

Problem Set 3 (5%) covers hash tables and graph traversal, and is due on **October 10**. Late submissions lose 10% per day.

The **Midterm Exam** (25%) will be held on the **second Tuesday of October** and covers all material through binary search trees. It is closed-book.

Quizzes (10%) are short, unannounced, and administered **weekly** during section — see the course calendar for section times.

The **Final Project** (30%) is a substantial implementation assignment covering the full course syllabus. Due date: **TBD** — will be announced after the midterm.

Class Participation (20%) is assessed continuously throughout the semester based on section attendance and engagement.

Course Schedule

Note: Problem Set 3 date below reflects a schedule revision made after the syllabus text above was drafted.

Week	Topic	Deliverable
1	Course Introduction, Big-O Review	--
2	Arrays and Dynamic Arrays	--
3	Linked Lists	Problem Set 1 due Friday
4	Stacks and Queues	--
5	Recursion and Divide-and-Conquer	--
6	Sorting Algorithms	Problem Set 2 due Friday
7	Midterm Review	--
8	Binary Search Trees	Midterm Exam
9	Hash Tables and Graphs	Problem Set 3 due Oct 17
10	Graph Traversal (BFS/DFS)	--
11	Shortest Paths	--
12	Dynamic Programming	--
13	Advanced Topics	--
14	Final Project Work Session	--
15	Final Project Presentations	Final Project due

Course Policies

Academic integrity is taken seriously; all submitted work must be the student's own. Extensions are granted only for documented emergencies. This section contains no deliverables or deadlines.